

Forest Rail

The Jingzhang Railway Heritage Park already left the rails in the city: people run, push strollers, walk along the tracks. This proposal does not reinvent that park, and does not turn it into a tech showcase. It asks something narrower — **how the Dazhongsi end connects into the park, where people go after exiting the metro, where they can stop, and whether they can sit down and work quietly** [source:AGENT-TASKBOOK].

The answer is Forest Rail:

- **Forest**: Treat the Dazhongsi section as a park — manicured lawn, spaced trees, canopy above rooftops. Look out and see trees, not the glass wall opposite.
 - **Rail (two tracks, separated)**:
 - **Decorative guiding rails**: rusted / replica rails / ground painting for direction only; people walk soft paving (timber, crushed stone). **Children do not run or climb on the rails.**
 - **AI operating rails (concept)**: passenger short-haul + automatic cargo configuration; physically separated from walking and decorative rails; subject to professional feasibility.
- > One line: buildings under trees, people in a park, rusted rails pointing the way, operating rails elsewhere — look out and see nature.

This proposal carries forward the Jingzhang Railway spirit of daring breakthrough — turning heritage rails into walkable, sittable, shared public infrastructure — without empty empowerment slogans or viral landmarks.

What this proposal refuses: district-wide building control, viral landmarks, light shows, planting strips posing as parks, private pods disguised as shared desks, decorative rails treated as climbing gear.

Only one rail-guided corridor is added (guiding rails primary; operating rails only on isolated conceptual segments). Outside it, the existing street network stays.

Design Basis and Source List

This proposal follows the Pre-qualification Announcement for the Centennial Jingzhang AI Innovation Belt Urban Design International Open Call [source:OFFICIAL-ANNOUNCEMENT], with machine-readable references from `brief/site-package/` [source:SITE-PACKAGE].

Current spatial data uses provisional boundaries (`geometry/site_boundary.geojson`, `geometry/key_areas.geojson`), labeled `provisional_constraint`, `official_boundary=false`. All metrics and drawings must be recalculated when official boundaries are released. This limitation does not block content scoring, but spatial conclusions in this text are for design discussion only.

The source registry (`data/source_registry.json`) lists 7 formal sources, 1 background source, and 1 provisional-only source. Background and provisional sources are not elevated to statutory authority [source:SOURCE-REGISTRY].

Authority order in this package: GeoJSON → metrics → matrices → sources/assumptions/self_check → proposal.md → figures → HTML → PDF. Concept renders do not replace evidence diagrams.



Three-Level Scope Framework

The proposal follows the three tiers specified in the announcement [source:SITE-PACKAGE]:

| Tier | Area | What Forest Rail does here |

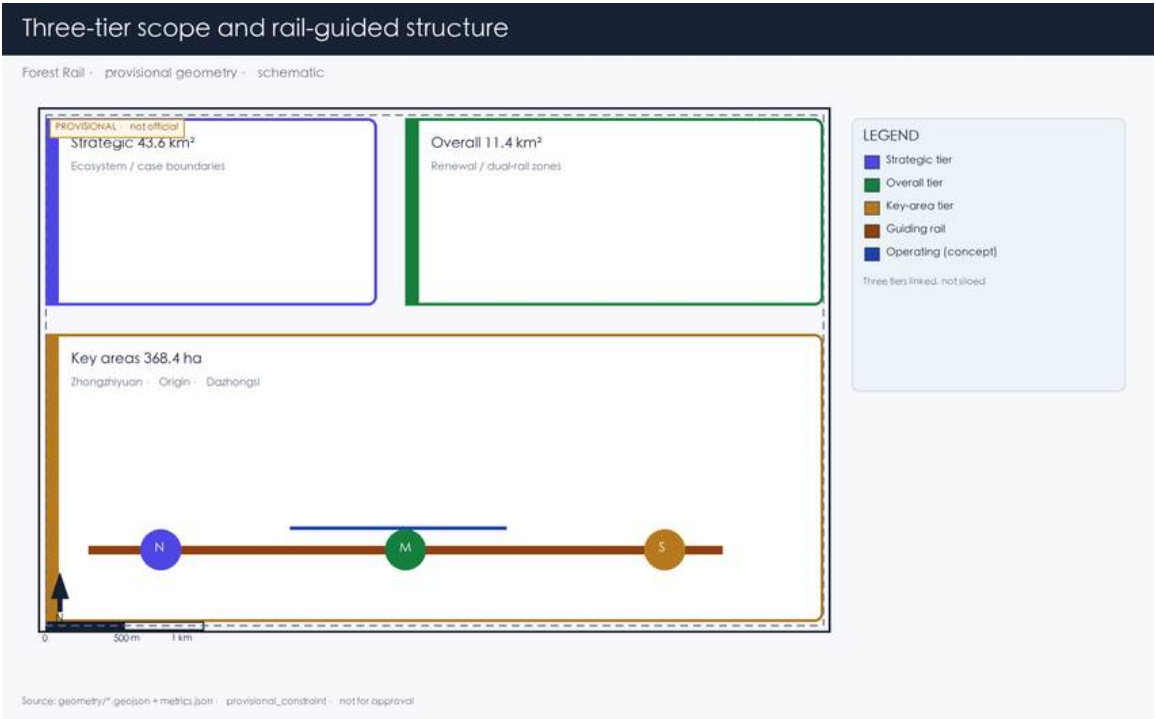
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| Strategic research (43.6 km²) | AI industry ecosystem and urban morphology | Confirms the Jingzhang corridor as spine; Forest Rail can extend north from Dazhongsi |

| Overall design (11.4 km²) | Urban renewal around the heritage park | Rail corridor links three key areas; park walking and soft paving; canopy as character control |

| Key area design (368.4 ha) | Three detailed design zones | Dazhongsi completes the park set: canopy, rail path, soft paving, shared desks, rust |

Tiers are linked: strategy sets direction, overall design lands layers, key areas test whether components fit parcels.



Coordinated Research Area: Industry and Future City Research

The Jingzhang Railway is the narrative already embedded here. It opened in 1909; after going underground in 2019, the surface alignment became a heritage park corridor. The announcement requires integrating centennial Jingzhang culture, Zhongguancun innovation culture, and AI innovation culture [source:AGENT-TASKBOOK].

Forest Rail's judgment is narrow: ****do not invent a new story — use the rails; do not invent a new green corridor — connect the park to everyday Dazhongsi.**** Rails stay on the ground and rust; people walk soft paving between them. The line connects three key areas — Zhongzhiyuan by Qinghe, Origin Community near universities, Dazhongsi by the metro. The rail-guided corridor is the spine, not decoration.

Industry links follow routes people actually walk:

- ****University sourcing****: work moves south along the heritage corridor, not floating circles
- ****Open-source collaboration****: Origin Community hosts release and code work
- ****Enterprise translation****: Dazhongsi hosts agents, terminals, and data-element firms
- ****Public services****: ask-a-person nodes along the rail path, not unmanned screen walls
- ****International visits****: walkable rail-path itineraries, not parade stages

Naming: Forest Rail is both title and spatial description. No logo needed — the track is the mark. Rust and canopy green are enough.

Urban Parks and Linear Parks: Global / Domestic Cases (Transfer Boundaries)

At the strategic tier, Forest Rail does not invent a new green-corridor story. It aligns transferable mechanisms from well-known urban and linear parks to the Jingzhang heritage corridor. Cases below are publicly built or operating projects. This proposal learns publicness and linear organization — not viral installations or climbable heritage consumption.

| Case | Transferable mechanism | Boundary for this proposal (learn / do not learn) |

|-----|-----|-----|

| New York Central Park | Open lawns, all-ages public realm, urban-forest scale | ****Learn****: treat Dazhongsi as a real park (lawn / spaced trees / seating), not a planting strip; ****Do not****: copy a full mega-park facility inventory |

| New York High Line | Abandoned elevated rail reused as a linear walking park | ****Learn****: railway heritage → walkable public space; ****Do not****: elevated tourist consumption; contrast — our decorative rails ≠ climbing equipment; people walk soft paving |

| Philadelphia Rail Park | Rail corridor opened in phases for everyday community use | ****Learn****: phased pilots; lightweight near-term starts; ****Do not****: market unfinished segments as finished experience |

| London King’s Cross public realm | Station-adjacent squares, walking, mixed-use stitching | ****Learn****: exit station into public space (Dazhongsi four quadrants); ****Do not****: mall-led plaza logic |

| Madrid Río | Continuous riverside park beside / over infrastructure | ****Learn****: blue-green continuity and walking priority over display; ****Do not****: mega covering-structure civil works |

| Seoul Cheonggyecheon / Seoulllo | Linear urban public space recovery; elevated rail segment as park | ****Learn****: linear publicness and everyday civic use; ****Do not****: massive channel-excavation narratives |

| Shanghai Xuhui Runway Park | Former airport runway → linear park with walking / cycling layers | ****Learn****: transport remnant translated into everyday linear park; ****Do not****: aviation-theme photo-booth stacking |

| Beijing Olympic Forest Park / Chaoyang Park | City-scale parks, all-ages programs, lawn and canopy | ****Learn****: urban-forest character and all-ages inclusion; ****Do not****: claim a new mega-park land take |

| Jingzhang Railway Heritage Park (existing) | Rails already in the city; everyday walking already happens | ****Learn****: connect into Dazhongsi daily life as incremental link; ****Do not****: redo the existing park or invent a parallel green corridor |

Stance: carry forward Jingzhang daring — move heritage rails from “to be seen” toward “to guide, to share work, to host conceptual short-haul” — with the breakthrough landing on safety separation and everyday publicness, not empowerment slogans or viral gear.

Park case background sources are registered in `sources.json` (not statutory). North American urban/linear parks: [source:CASE-CENTRAL-PARK], [source:CASE-HIGH-LINE], [source:CASE-RAIL-PARK]. Europe and East Asia: [source:CASE-MADRID-RIO], [source:CASE-CHEONGGYECHEON]. Domestic and local baseline: [source:CASE-XUHUI-RUNWAY], [source:CASE-OLYMPIC-FOREST], [source:CASE-JZ-HERITAGE-PARK].

Overall Design: Rail-Guided Corridor and Urban Renewal

Corridor Structure

The rail-guided corridor follows the Jingzhang heritage park alignment, with Dazhongsi as the southern gateway. It is not a newly drawn boundary — it reuses the railway alignment. ****The park is already walked; this proposal mainly connects Dazhongsi Station into that park.****

...

Zhongzhiyuan (N) —rail— AI Origin Community (M) —rail— Dazhongsi (S)

| |

Qinghe interface Campus-district walking link

...

Tracks along the corridor serve two purposes and are ****physically separated**** [source:AGENT-TASKBOOK]:

| Purpose | Material | Function | Boundary |

|-----|-----|-----|-----|

| ****Decorative guiding rails**** | Rusted / replica rails / ground painting + soft paving | Direction: eyes follow rust; feet on timber or crushed stone; sittable, slow | No running or climbing on rails; no light strips; signage minimal (Mode A) |

| ****AI operating rails (concept)**** | Isolated operable segments | Low-speed passenger short-haul + automatic cargo/materials configuration | Separated from walking and decorative rails; not commercially mature; subject to professional study |

Most stretches are decorative guiding rails only. A few isolated stretches conceptually retain AI operating capacity. Pedestrians have priority at crossings; operating rails yield. If operating rails fail or are never deepened, guiding rails and the park still stand on their own.

Land Use, Building Scale, and Retain-Renovate-Demolish Strategy

Renewal within the overall design scope falls into three categories:

- ****Retain****: Existing quality buildings along the heritage corridor — unchanged
- ****Renovate****: Buildings needing ground-floor program and facade adjustment — remove LED screens, increase under-canopy transparency
- ****Renew****: New construction on identified low-efficiency sites — building height controlled below tree canopy

Land use classification follows national standards [standard:MNR-LAND-USE-CLASSIFICATION-GUIDE], covering the design boundary without overlap [data:geometry/land_use.geojson#LU-001]. Building footprints and retain/renovate/renew classification are recorded in [data:geometry/buildings.geojson#BLDG-001].

Where FAR, building height, or road setbacks lack official data, they are marked as unknown. No numbers are fabricated.

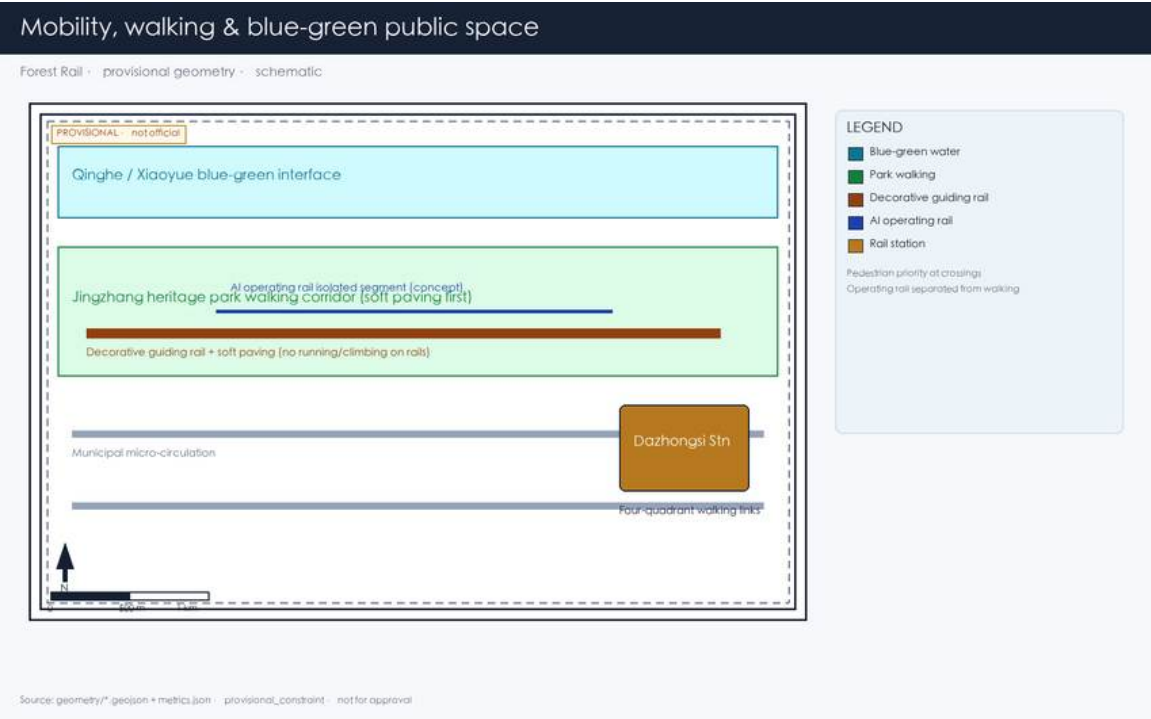
Transport, Rail, Municipal Infrastructure, and Public Services

Transport responds to the announcement's requirements for rail station integration, road micro-circulation, walking network gaps, and green transport [depth:traffic_rail_slow_parking].

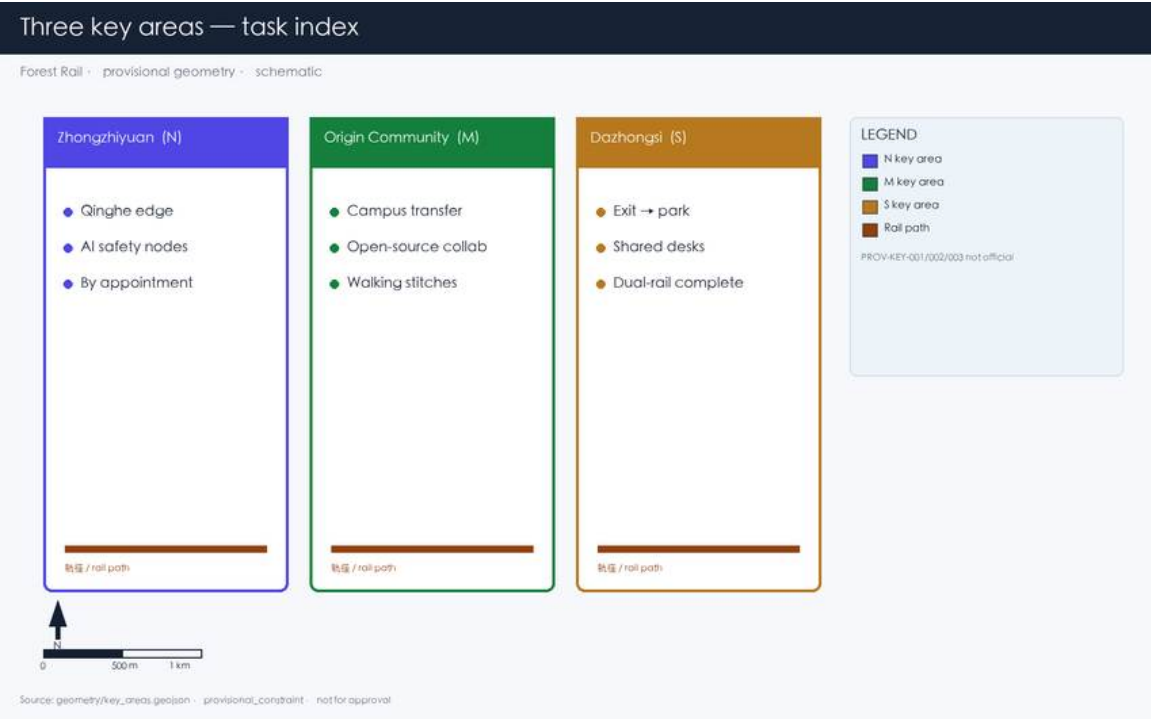
Key interventions:

- **Dazhongsi Station four-quadrant pedestrian connection****: Metro Line 13 Dazhongsi Station — establish walking connections across all four quadrants of the intersection, with rail paths extending from station exits into the district
- **North Fifth Ring crossing node****: Pedestrian linkage where the heritage corridor crosses the North Fifth Ring Road (underpass or overhead facility) to maintain north-south continuity
- **Rail path / municipal road intersections****: Pedestrians have priority at crossings; rail tracks embedded in road surface indicate direction; no traffic signals govern the rail path

Road and walking network layers remain within the submission boundary [data:geometry/roads.geojson#ROAD-001], cross-checked with public space and green space. With a provisional boundary, transport conclusions serve as design discussion only.







All three key areas appear in `geometry/key\_areas.geojson`. Current data is provisional constraint; the proposal, HTML, and self_check state that it cannot serve as approval or scoring basis.

A Day Here: Personas and Scenarios

Five User Personas

| Persona | Who | What they do on Forest Rail | Data boundary |

|-----|-----|-----|-----|

- | ****Walking commuter programmer**** | AI engineer living nearby, walks to work daily | Exits Dazhongsi Station, walks 800m along the rail path to the office; eats lunch under the canopy; walks back along the rail path after work | No personal commute tracking; rail path foot traffic is anonymous counting only |
- | ****Parent walking with child**** | Resident of surrounding neighborhoods | Pushes a stroller along the soft paving in the evening; child runs on soft paving beside the rails (not on the rusted rails); sits on under-canopy benches chatting | No family profiling; no commercial push notifications |
- | ****Corporate visitor**** | Business guest of a major company | Arrives at Dazhongsi Station, rail path guides to company showroom; walks along rail path to next node after the meeting | Company logos and cases require rights clearance |
- | ****Solo freelancer**** | Person without a fixed office | Finds an open shared desk in the park workstation precinct; opens when quiet, closes when crowded; hot-desk, no reserved seats | No recording of workstation user identity |
- | ****Retired daily walker**** | 70-80 years old, lives nearby | Walks along the rail path every morning, sits on the bench at the curve watching trees; pavement must be non-slip | No health data collection |

Scenario Cards (≥10)

| # | Scenario | Spatial carrier | What happens | Who uses it |

|---|-----|-----|-----|-----|

- | 01 | Exit station, enter park | Dazhongsi Station exits | Decorative guiding rails point to park entries; canopy provides auxiliary shade; rain relies on shelters/arcades as backup — not light shows | All arrivals |
- | 02 | Rail path walking | Entire Dazhongsi rail path | Eyes follow rusted rails; feet on timber or crushed stone. Speed naturally slows. No destination signs urging you forward | Commuters, walkers |
- | 03 | Shared under-canopy workstations | Dedicated park precinct with glass-partitioned multi-desk bays | Shared hot-desks: sit if free. Opens when quiet (noise <55dB, no extreme heat, no heavy rain). AI manages open/close by time and environment | Freelancers, lunch-breakers, passersby working briefly |
- | 04 | Rusted rail curve bench | Rail path turning points | Benches on the outside of curves; sit and watch the rusted rail and trees. No signage; you discover it naturally by sitting down | Elderly, parents with children |
- | 05 | Company showroom along the rail | Dazhongsi ground-floor retail | AI agent and terminal company display spaces open toward the rail path; visible as you walk past | Visitors, passersby |
- | 06 | Rainy day under-canopy shelter | Dense canopy segments + node shelters | Canopy as auxiliary shade; heavy rain backed up by artificial shelters / semi-open arcades; AI prompts open seating and shelter nodes | Everyone |
- | 07 | Open-source launch hall | Origin Community | Code contribution display and small presentations for universities and open-source communities. Physical screen scrolling real-time commits | Developers, university faculty and students |
- | 08 | Inter-node AI operating rail (concept) | Isolated operable segments | Conceptual low-speed passenger short-haul + automatic cargo pods linking Zhongzhiyuan, Origin Community, and Dazhongsi; separated from walking and decorative guiding rails | Concept verification stage, not mixed daily use |
- | 09 | Night walking | Entire rail path | Night lighting is minimal: ground-embedded low-illuminance lights mark soft paving edges only. No light shows, no projections | Night runners, late returners |
- | 10 | Qinghe innovation corridor | Zhongzhiyuan along Qinghe | Waterfront walkway connects to the rail path, stormwater grass slopes double as resting lawns. AI safety testing nodes open by appointment | Visitors, walkers |
- | 11 | Data element salon | Dazhongsi district | Under compliance, authorization, and auditability, displays data element and digital asset circulation as an urban service interface | Enterprise clients, regulators |
- | 12 | University tech transfer street | Origin Community | Incubation, display, legal, IP, and investment services organized along the rail path for university technology transfer | Startup teams, investors |

All AI governance follows data minimization, open sourcing, explainability, and human review. Urban intelligent agents assist in identifying walking network gaps, public space activity, and facility maintenance needs but do not replace planning approvals or output unauthorized personal profiles.

Three Quiet Landmarks (Not Attractions)

No "check-in points" are designated here. The following are places you notice while walking — no signage, no wayfinding, no photo-taking recommendations:

- **Rusted rail curve with an old Scholar Tree****: At a bend in the rail path, an existing large Scholar Tree. The curve makes you turn your head and see the tree. Unnamed, unmarked.
- **Rail path–Xiaoyue River junction, crushed stone square****: Where the rail path crosses Xiaoyue River, a crushed stone surface forms a small square. Water, rusted rail, and canopy meet here. No sculpture.
- **Old platform retaining wall****: If Jingzhang Railway-era platform remnants exist in the district (retaining walls, steps), they are kept as found. The wall surface is not cleaned; moss grows. A bench is placed beside it.

Renewal Project List and Phasing

| ID | Project | Type | Key dependencies | Phase |

|---|-----|-----|-----|-----|

- | FR-01 | Dazhongsi Station four-quadrant rail path connection | Walking / rail integration | Rail station, intersection traffic organization | Near-term pilot |
- | FR-02 | Rail path wayfinding segment installation (Dazhongsi) | Public space | Rusted rail sourcing, soft paving materials, drainage | Near-term pilot |
- | FR-03 | Canopy layer planting (Dazhongsi) | Greening | Tree specification, soil improvement, utility avoidance | Near-term start, 3-5 year canopy maturity |
- | FR-04 | Under-canopy workspace ground floor renovation | Building renovation | Ownership, ground-floor program adjustment | Medium-term |
- | FR-05 | Zhongzhiyuan Qinghe innovation interface | Blue-green / industry | River blue line, ecology and flood control conditions | Medium-term |
- | FR-06 | Origin Community campus walking integration | Walking | Campus boundary, wall removal negotiation | Medium-term |
- | FR-07 | Inter-node AI operating rail concept segment (passenger short-haul + automatic cargo) | New infrastructure (concept) | Engineering feasibility, safety assessment, physical separation | Long-term / pending |
- | FR-08 | Full rail path connection (three key areas linked) | Public space / walking | North Fifth Ring crossing, corridor ownership | Long-term |

Near-term pilots can start with lightweight interventions (spread crushed stone, place benches, plant saplings) without waiting for all engineering conditions. Medium and long-term projects require confirmed regulatory plans, municipal engineering, and property rights.

Phasing spatial evidence [data:geometry/phasing.geojson#PHASE-001].

Metrics Framework

Metrics are divided into three categories:

| Type | Examples | Data source | Precision |

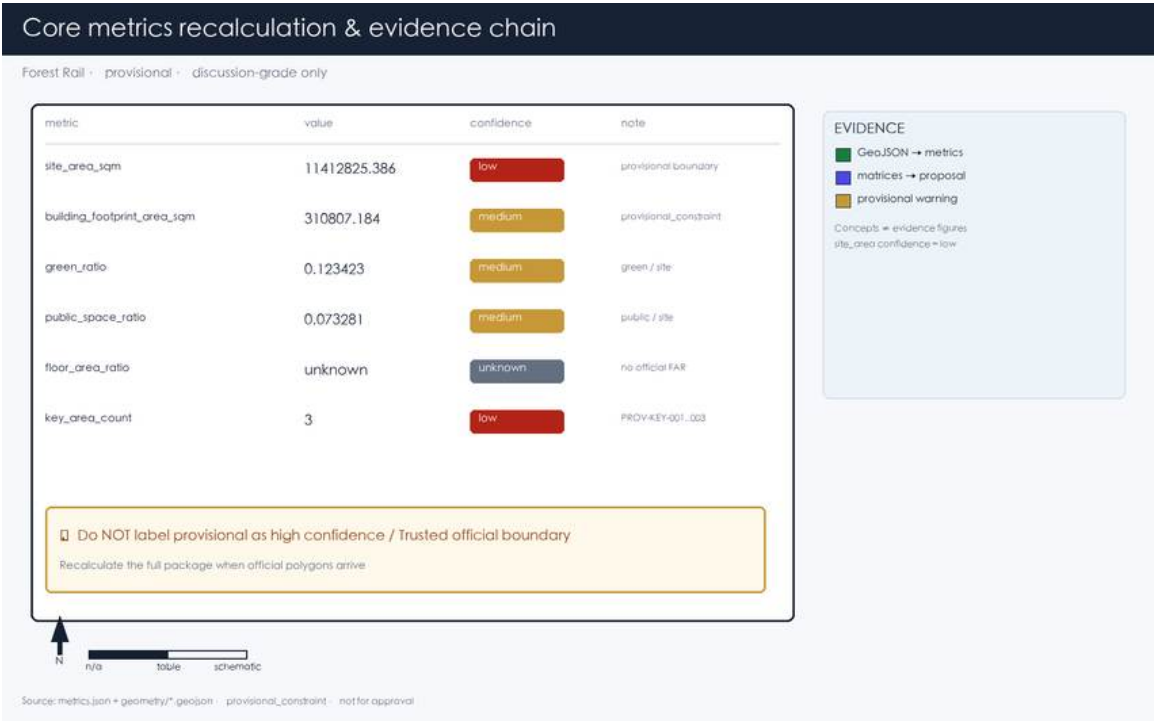
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| Spatial metrics calculable from submitted geometry | Boundary area, green ratio, public space ratio, building footprint | GeoJSON layers | Limited by provisional boundary |

| Control metrics requiring official regulatory plans | FAR, building height, building density, setbacks | Pending official plans | Currently marked unknown |

| Performance metrics requiring ongoing operational data | Rail path daily foot traffic, under-canopy workstation utilization, node access frequency | Post-operation collection | Currently design reference values |

Three categories enter `metrics.json`, `assumptions.json`, and `compliance\_matrix.json` respectively. Full calculations are in `metrics.json`; this text explains design intent only.



Key spatial metrics can be verified against [metric:site_area_sqm] and [data:geometry/green_space.geojson#GREEN-001].

Compliance Matrix Coverage

`compliance\_matrix.json` maps each requirement from Announcement 1.3-1.5 and agent.1-agent.6:

| Requirement | Proposal section | Core evidence |

|-----|-----|-----|

| 1.3 World-class AI innovation ecosystem | Strategic research + Key areas | Innovation chain spatial layout, scenario cards |

| 1.4 Three-tier scope | Three-tier scope | Task table per tier |

| 1.5.3.1 Zhongzhiyuan | Key areas: Zhongzhiyuan | key_areas PROV-KEY-001 |

| 1.5.3.2 Origin Community | Key areas: Origin Community | key_areas PROV-KEY-002 |

| 1.5.3.3 Dazhongsi | Key areas: Dazhongsi | key_areas PROV-KEY-003 |

| agent.1 Overall concept | Overall design: Corridor structure | Rail-guided corridor + three key area linkage |

| agent.2 AI full-stack innovation | Strategic research + Zhongzhiyuan | Innovation chain, safety governance nodes |

| agent.3 AI+ scenarios | Personas and scenarios | 12 scenario cards |

| agent.4 Public space and landmarks | Dazhongsi urban forest + quiet landmarks | Five components + three quiet landmarks (not attractions) |

| agent.5 Cultural narrative integration | Strategic research: Why rails | Jingzhang Railway to rusted rail to Forest Rail |

| agent.6 Global activities and long-term operations | Renewal projects: Phasing | Near-term lightweight to medium-term renewal to long-term connection |

Risk, Copyright, and Compliance

Provisional Boundary

All spatial conclusions are limited by provisional boundary status. When official boundaries are released: re-run scaffold, self-check, drawings, and HTML. Single-file replacement is not sufficient.

AI Operating Rail Concept (Passenger Short-Haul + Automatic Cargo)

AI operating rails are a conceptual suggestion, not an engineering feasibility conclusion: low-speed passenger short-haul plus automatic cargo/materials pods. Rail operations on the Jingzhang heritage / dense urban road network cannot be described as commercially mature. Proposal language: conceptual suggestion, subject to professional study; physically separated from heritage preservation, decorative guiding rails, and pedestrian space; decorative rails forbid climbing and running.

Missing Data

Gaps listed in `missing\_data\_checklist.csv` — official boundaries, regulatory plans, roads, municipal engineering, heritage protection — enter `assumptions.json` and self-check. Conclusions lacking official conditions are downgraded to pending confirmation.

Copyright

The proposal provides bilingual versions (this file and `proposal.md`). Images, data, and code assets are documented in `sources.json` and `report/copyright\_statement.md`. HTML pages do not load remote scripts, map tiles, fonts, iframes, or external APIs.

Disclaimer

This proposal does not claim official approval, regulatory plan status, final property rights, final construction scale, or implementation guarantee.

References

See [source:OFFICIAL-ANNOUNCEMENT], [source:AGENT-TASKBOOK], [source:SITE-PACKAGE], `sources.json`, `metrics.json`, `compliance\_matrix.json`.

- brief/public-brief.md

- brief/site-package/design_brief.json

- brief/site-package/allowed_design_space.json

- data/processed/agent_fact_pack.md

- data/processed/project_scope_summary.csv

- data/processed/agent_task_requirements.csv

- data/processed/missing_data_checklist.csv

- Full source index: `sources.json`, `metrics.json`, `compliance\_matrix.json`, `standard\_matrix.json`, `design\_depth\_matrix.json`